

**PARUL INSTITUTE OF ENGINEERING & TECHNOLOGY**  
PARUL UNIVERSITY, VADODARA

**DEPARTMENT OF ELECTRONICS AND COMMUNICATION  
ENGINEERING**

**OEP PROJECT REPORT**  
ON

**MULTI-ZONE INDUSTRIAL SAFETY SYSTEM WITH  
INTERLOCK AND ENVIRONMENTAL MONITORING**

ANALOG CIRCUITS LAB (303107254)

**SUBMITTED BY**

Mayur Patil — Enrollment No: 2503031070026  
Pratik Kshirsagar — Enrollment No: 2503031070016  
Sanika Shende — Enrollment No: 2503031070051  
Prem Dawale — Enrollment No: 2503031070042  
Chaitanya Motba — Enrollment No: 2503031070045  
Rudraksh Ahire — Enrollment No: 2503031070002

**GUIDED BY**

**Prof. Mithilesh Yadav**  
Department of Electronics & Communication Engineering

**Academic Year: 2025–2026**

# **CERTIFICATE**

## **DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING**

**PARUL INSTITUTE OF ENGINEERING & TECHNOLOGY  
PARUL UNIVERSITY, VADODARA**

This is to certify that the project titled "MULTI-ZONE INDUSTRIAL SAFETY SYSTEM WITH INTERLOCK AND ENVIRONMENTAL MONITORING" has been successfully completed by:

Mayur Patil — Enrollment No: 2503031070026  
Pratik Kshirsagar — Enrollment No: 2503031070016  
Sanika Shende — Enrollment No: 2503031070051  
Prem Dawale — Enrollment No: 2503031070042  
Chaitanya Motba — Enrollment No: 2503031070045  
Rudraksh Ahire — Enrollment No: 2503031070002

They are bonafide students of the Third Semester, Bachelor of Engineering in Electronics and Communication Engineering at Parul Institute of Engineering & Technology, affiliated to Parul University, Vadodara, during the academic year 2025–26.

This project has been completed in partial fulfilment of the course requirements for the award of the degree of Bachelor of Engineering in Electronics and Communication Engineering, as prescribed by Parul University. It is certified that all corrections and suggestions indicated during internal assessment have been duly incorporated in the report. The project report has been approved as it satisfies the academic requirements for the prescribed degree.

Signature of Internal Guide: \_\_\_\_\_

**Prof. Mithilesh Yadav**

Assistant Professor, Department of ECE

Signature of Head of Department: \_\_\_\_\_

**Prof. Kalpesh Jadav**

Head of Department, ECE

Date: \_\_\_\_\_

## **ACKNOWLEDGEMENT**

The successful completion of this project has been made possible through the support, guidance, and encouragement of several individuals, and we take this opportunity to express our heartfelt gratitude to all of them.

We are deeply grateful to our project guide, Prof. Mithilesh Yadav, Department of Electronics and Communication Engineering, Parul University, for the invaluable guidance, continuous motivation, and constructive feedback provided throughout the course of this project. The technical insights and practical perspective shared by our guide greatly aided us in designing and implementing the system effectively.

We express our sincere thanks to Prof. Kalpesh Jadav, Head of the Department of Electronics and Communication Engineering, for providing us with the necessary laboratory facilities, resources, and a supportive academic environment to carry out this work.

We also acknowledge the support of the laboratory staff and technical assistants whose assistance during hardware implementation and testing was invaluable.

Finally, we thank our families and friends for their constant encouragement and moral support during the course of this project.

Regards,

Mayur Patil — Enrollment No: 2503031070026

Pratik Kshirsagar — Enrollment No: 2503031070016

Sanika Shende — Enrollment No: 2503031070051

Prem Dawale — Enrollment No: 2503031070042

Chaitanya Motba — Enrollment No: 2503031070045

Rudraksh Ahire — Enrollment No: 2503031070002

## **ABSTRACT**

Industrial safety is a critical requirement in modern manufacturing and processing environments, where human lives and equipment are continuously exposed to various physical and environmental hazards. Conventional safety systems are often limited to single-zone monitoring with minimal integration, leaving critical gaps in real-time hazard detection and access control.

This project presents a Multi-Zone Industrial Safety System with Interlock and Environmental Monitoring, designed and implemented using an Arduino Mega 2560 microcontroller as the central processing unit. The system is capable of monitoring two independent safety zones — Zone A and Zone B — simultaneously, using dedicated RFID-based access control mechanisms at each entry point.

Zone A is equipped with a servo motor-controlled gate and an RFID reader that grants or denies access based on card authentication. Zone B employs a stepper motor-driven gate along with a secondary RFID reader for personnel verification. Both zones are continuously monitored using a DHT11 temperature and humidity sensor and an MQ-series gas sensor that detect abnormal environmental conditions such as elevated temperature or hazardous gas concentration.

The interlock mechanism forms the core safety feature of this system. When an unsafe condition is detected in either zone — such as high temperature or dangerous gas levels — the interlock immediately blocks all entry while simultaneously permitting personnel already inside to exit safely through an emergency exit button. An LCD display with I2C interface provides real-time status information, while a buzzer and LED indicators provide audio-visual alerts.

The system was implemented on both a breadboard prototype and a PCB for validation. Experimental results confirmed that the system correctly enforces access control, triggers alerts under hazardous conditions, and maintains fail-safe interlock behavior. The project demonstrates a practical, low-cost, and reliable approach to industrial zone safety management with significant educational and real-world applicability.

## **TABLE OF CONTENTS**

|                                                           |    |
|-----------------------------------------------------------|----|
| Chapter 1: Introduction                                   | 1  |
| Chapter 2: Objectives                                     | 3  |
| Chapter 3: Importance, Advantages, and Limitations        | 4  |
| Chapter 4: Components Used                                | 6  |
| Chapter 5: Block Diagram Explanation                      | 9  |
| Chapter 6: Circuit Description                            | 11 |
| Chapter 7: Implementation                                 | 13 |
| Chapter 8: Working Principle                              | 16 |
| Chapter 9: Applications                                   | 18 |
| Chapter 10: Observations                                  | 20 |
| Chapter 11: Conclusion                                    | 21 |
| Chapter 12: Future Scope                                  | 22 |
| Chapter 13: Sustainable Development Goals (SDG Alignment) | 24 |

## **CHAPTER 1: INTRODUCTION**

This chapter provides the background and context that motivated the development of the Multi-Zone Industrial Safety System. It outlines the core problem addressed, describes the proposed solution, and defines the scope of the project.

### **1.1 Background**

Industrial environments — including chemical plants, manufacturing units, warehouses, and power generation facilities — are inherently hazardous. Workers in such environments are continuously exposed to risks such as toxic gas leakage, high-temperature zones, unauthorized personnel intrusion, and restricted zone violations. A single safety lapse in such facilities can lead to catastrophic consequences, including injuries, fatalities, equipment damage, and production losses.

Traditional industrial safety practices often rely on manual monitoring, basic alarm systems, and physical barriers. However, these approaches are limited in their response speed, accuracy, and integration capability. Modern industry demands intelligent, automated, and multi-functional safety systems that can monitor multiple zones, enforce access control, detect environmental hazards, and respond instantly without human intervention.

### **1.2 Problem Statement**

Most small and medium-scale industries lack integrated safety systems that can simultaneously handle access control and environmental monitoring across multiple zones. Single-point safety systems fail to provide comprehensive protection, and manual monitoring is prone to human error and delayed response. There is a clear need for an affordable, automated, and reliable multi-zone safety system that can prevent unauthorized access, detect environmental hazards in real time, and enforce interlocking mechanisms to protect personnel.

### **1.3 Proposed System Overview**

This project proposes and implements a Multi-Zone Industrial Safety System using the Arduino Mega 2560 microcontroller as the central controller. The system integrates two RFID-based access control zones, environmental sensors for temperature and gas monitoring, motor-driven gate control, an interlock mechanism, and real-time display and alert systems. The entire system is designed to be cost-effective, easy to deploy, and scalable to larger industrial environments.

Zone A represents a general-access industrial entry point where a servo motor-controlled gate regulates personnel entry based on RFID card verification. Zone B represents a restricted or hazardous area entry point where a stepper motor-driven gate provides more precise and controlled access. Both zones share a common environmental monitoring network that continuously assesses temperature, humidity, and gas concentration levels. If any parameter exceeds a predefined safety threshold, the interlock system activates immediately — preventing further entry while enabling safe evacuation through an emergency exit button.

#### **1.4 Scope of the Project**

The scope of this project covers the design, hardware implementation, and testing of a two-zone industrial safety system. The system is intended for laboratory demonstration and educational purposes, with a design architecture that can be directly scaled to real-world industrial deployments. The project encompasses hardware circuit design, embedded programming using Arduino, sensor integration, motor control, display interfacing, and safety logic implementation.

## CHAPTER 2: OBJECTIVES

The primary objective of this project is to design and implement a reliable, multi-zone industrial safety system that integrates access control with environmental monitoring and a fail-safe interlock mechanism. The specific objectives of this project are as follows:

- To design and implement a dual-zone access control system using RFID technology, where each zone operates independently with its own reader and gate control mechanism.
- To integrate environmental monitoring sensors — specifically the DHT11 for temperature and humidity, and the MQ-series sensor for gas detection — to continuously assess safety conditions within the monitored zones.
- To implement an interlock mechanism that automatically blocks entry into any zone when environmental parameters exceed predefined safety thresholds, while simultaneously enabling personnel already inside to exit safely.
- To interface a servo motor for gate control in Zone A and a stepper motor for gate control in Zone B, demonstrating the use of different actuator types within the same safety system.
- To provide real-time status feedback to operators using a 16×2 LCD display with I2C interface, displaying zone status, access results, and environmental readings.
- To incorporate audio-visual alert systems using a buzzer and LED indicators for immediate notification of unsafe conditions or unauthorized access attempts.
- To provide a dedicated emergency exit button that enables safe evacuation of personnel regardless of the current system state.
- To validate the complete system through hardware testing and Proteus-based simulation, confirming that all functional and safety requirements are met.
- To align the project design with relevant Sustainable Development Goals, highlighting the broader societal impact of the system.

## **CHAPTER 3: IMPORTANCE, ADVANTAGES, AND LIMITATIONS**

This chapter discusses the significance of the proposed system within the broader context of industrial safety engineering, enumerates its principal advantages over conventional approaches, and candidly identifies its current limitations.

### **3.1 Importance of the System**

Industrial safety management is no longer a regulatory checkbox — it is an operational necessity. With global industrial accidents causing thousands of fatalities and billions of dollars in economic losses every year, the integration of automated safety systems has become essential. The Multi-Zone Industrial Safety System with Interlock and Environmental Monitoring addresses this need by providing a practical, low-cost, and reliable solution that can be deployed across a wide range of industrial settings.

The system is particularly important for the following reasons. First, it provides continuous, 24/7 monitoring without dependence on human attention, eliminating the risk of delayed response due to human error or fatigue. Second, the dual-zone architecture allows simultaneous protection of multiple entry points, making it suitable for facilities with distinct restricted and general access areas. Third, the interlock mechanism ensures that safety is enforced proactively rather than reactively — blocking entry before a person is exposed to a hazardous environment, rather than sounding an alarm after the fact. Fourth, the emergency exit feature reflects a humanitarian design philosophy: even in a lockdown, personnel inside are never trapped.

From an academic perspective, this project serves as an excellent demonstrator of embedded systems integration, combining microcontroller programming, sensor interfacing, motor control, and real-time decision-making within a single cohesive system.

### **3.2 Advantages of the System**

- **Multi-zone coverage:** The system monitors two independent zones simultaneously, providing comprehensive area-wide safety coverage from a single controller.
- **Automated access control:** RFID-based authentication removes the need for manual verification, reducing the risk of human error in access management.
- **Proactive hazard prevention:** The interlock mechanism prevents personnel from entering hazardous zones before they are exposed to danger, rather than alerting them after entry.
- **Fail-safe emergency exit:** The emergency exit button ensures personnel can always evacuate safely, regardless of system state or power conditions.
- **Real-time monitoring and alerts:** Continuous sensor readings with instant audio-visual alerts ensure that operators are immediately informed of any safety concern.

- Low cost and easy implementation: The system uses widely available, affordable components such as Arduino, RFID modules, and standard sensors, making it accessible for small and medium industries.
- Scalable architecture: The modular design allows additional zones, sensors, or actuators to be added without significant redesign of the core system.
- Educational value: The project demonstrates the integration of multiple electronics and embedded systems concepts in a real-world safety application.

### **3.3 Limitations of the System**

- Limited number of zones: The current implementation supports two zones. Expanding to more zones would require additional hardware and firmware modification.
- Basic sensor accuracy: The DHT11 sensor provides moderate accuracy ( $\pm 2^{\circ}\text{C}$  for temperature). For precision industrial environments, higher-grade sensors would be required.
- No remote monitoring: The current system operates locally without network connectivity. Remote monitoring and alerting are not available in this version.
- Fixed RFID database: The authorized RFID card list is stored in the microcontroller's memory. Dynamic addition or removal of authorized cards requires reprogramming.
- Single controller dependency: The system relies on a single Arduino Mega. A hardware fault in the controller could disable the entire system.
- No data logging: The system does not record access events, sensor readings, or alarm histories for audit purposes.

## CHAPTER 4: COMPONENTS USED

The following components were selected and used in the design and implementation of the Multi-Zone Industrial Safety System. Each component plays a specific and essential role in the overall functioning of the system.

### 4.1 Arduino Mega 2560 (Microcontroller)

The Arduino Mega 2560 is an open-source microcontroller board based on the ATmega2560 chip. It features 54 digital input/output pins, 16 analog input pins, 4 hardware UART serial ports, and 256 KB of flash memory. In this project, the Arduino Mega serves as the central controller, managing all input signals from sensors and RFID modules, executing the safety logic and interlock conditions, and generating output signals to control gates, displays, and alert devices. The Mega was specifically chosen over smaller boards like the Arduino Uno because of its multiple serial ports — essential for interfacing two independent RFID modules simultaneously — and its larger memory for handling the complete program logic.

### 4.2 RFID Module — MFRC522 (×2)

The MFRC522 is a highly integrated RFID reader/writer module that operates at 13.56 MHz and communicates with the microcontroller via SPI (Serial Peripheral Interface). Each module can read RFID cards or tags and return a unique identification number (UID). In this project, two MFRC522 modules are used — one for Zone A and one for Zone B. The Arduino Mega checks the UID of the presented card against a stored list of authorized UIDs. If the card is recognized, the corresponding gate opens; if not, access is denied and an alert is triggered. The modules are individually connected to the SPI bus with separate Slave Select (SS) pins to allow independent communication.

### 4.3 Servo Motor (Zone A Gate Control)

A servo motor is a rotary actuator that allows precise angular positioning through PWM (Pulse Width Modulation) signals. In this project, a standard 5V servo motor (such as the SG90 or MG90S) is used to control the gate in Zone A. When an authorized RFID card is presented and the interlock condition is not active, the servo rotates to 90 degrees to open the gate and returns to 0 degrees to close it after a defined delay period. The use of a servo motor in Zone A reflects a lightweight gate application where speed and simplicity are prioritized.

### 4.4 Stepper Motor with ULN2003 Driver (Zone B Gate Control)

A stepper motor is a brushless DC motor that moves in discrete, equal steps in response to digital pulses. It provides precise position control without the need for feedback sensors. In this project, a 28BYJ-48 stepper motor with a ULN2003 driver module is used for gate control in Zone B. The stepper motor is preferred here because Zone B represents a more controlled or restricted access point where gradual and precise gate movement is required. The driver module handles the high-current switching required to drive the motor coils from Arduino logic-level signals.

#### **4.5 DHT11 Temperature and Humidity Sensor**

The DHT11 is a basic digital sensor that measures ambient temperature (0°C to 50°C,  $\pm 2^\circ\text{C}$  accuracy) and relative humidity (20% to 80%,  $\pm 5\%$  accuracy). It communicates using a single-wire digital interface. In this project, the DHT11 monitors the temperature within the monitored zones. If the temperature exceeds a predefined safety threshold, the system identifies an unsafe condition and activates the interlock, blocking further entry into the zones. The DHT11 was selected for its simplicity, low cost, and easy integration with the Arduino platform.

#### **4.6 MQ-Series Gas Sensor**

The MQ-series gas sensors are analog sensors designed to detect specific gases in the surrounding environment. The MQ-2 sensor, used in this project, is sensitive to smoke, LPG, butane, methane, and hydrogen. The sensor outputs an analog voltage proportional to gas concentration. The Arduino reads this analog value and compares it against a predefined threshold. If the gas concentration exceeds the safe limit, the system triggers the interlock and raises an alert. Gas detection is critical in industrial environments where fuel leaks or combustible gases pose explosion and asphyxiation risks.

#### **4.7 LCD Display with I2C Interface (16×2)**

A 16×2 LCD (Liquid Crystal Display) is a character display capable of showing two rows of 16 characters each. The I2C module (typically based on the PCF8574 I2C expander) reduces the pin requirement from 6 to 2 (SDA and SCL), which is highly beneficial given the large number of devices connected to the Arduino Mega in this project. The LCD displays real-time information including zone access status, sensor readings, interlock state, and alert messages.

#### **4.8 Buzzer**

A 5V active buzzer is used as an audio alert device. In this project, the buzzer sounds when an unauthorized RFID card is detected, when the interlock is activated due to an unsafe condition,

or when the emergency exit button is pressed. The buzzer provides an immediate and unmistakable auditory warning that complements the visual alerts from the LED indicators.

#### **4.9 LED Indicators**

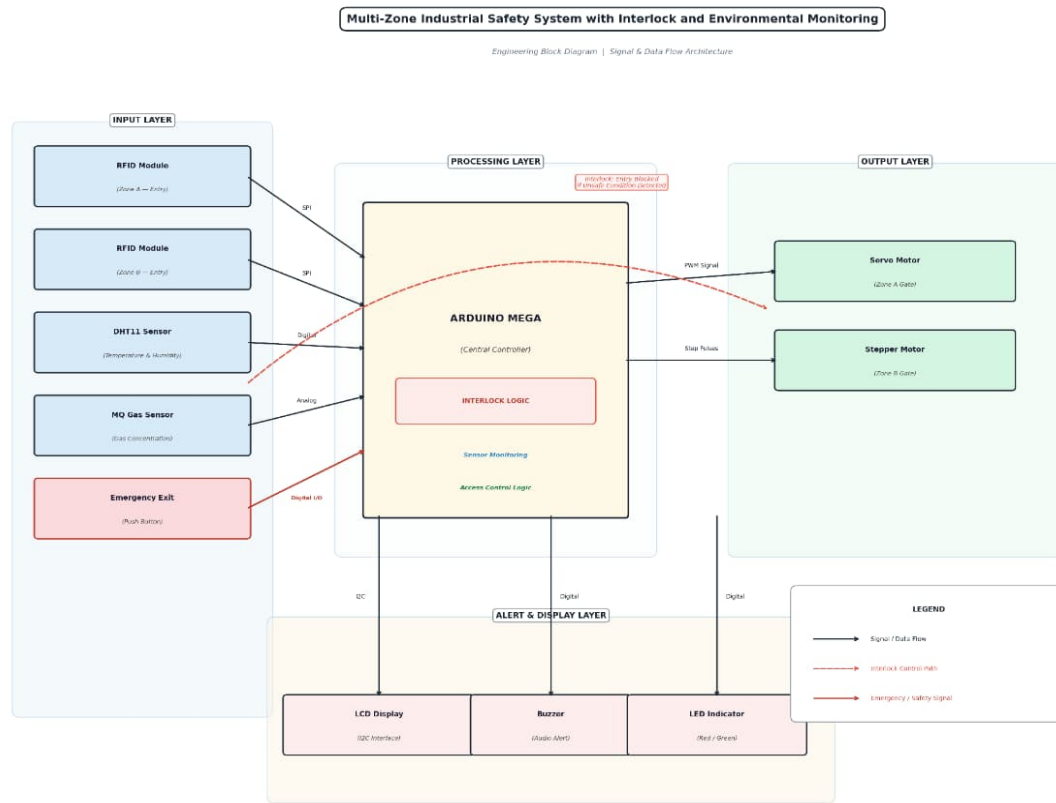
Red and green LEDs are used to provide clear visual status indicators. A green LED signals that the system is in a normal, safe state and access is permitted. A red LED signals a denied access event, an unsafe environmental condition, or an active interlock. LEDs are connected to digital output pins of the Arduino through appropriate current-limiting resistors ( $220\ \Omega$ ) to prevent damage.

#### **4.10 Emergency Exit Button**

A push button switch is connected to a dedicated digital input pin of the Arduino and configured as the emergency exit button. When pressed, the system overrides the interlock for exit purposes only — allowing gates to open for exit while still blocking entry. The button is placed in a normally open (NO) configuration and uses the Arduino's internal pull-up resistor for reliable operation.

## CHAPTER 5: BLOCK DIAGRAM EXPLANATION

This chapter presents and explains the block diagram of the Multi-Zone Industrial Safety System, illustrating the functional interconnections between all subsystems. The diagram serves as the architectural blueprint of the overall system and is essential for understanding the data and signal flow between components.



**Fig. 1: Block Diagram of Multi-Zone Industrial Safety System**

As illustrated in Fig. 1, the system is organized into five primary functional layers: the Input Layer, the Processing Layer, the Monitoring Layer, the Output/Actuator Layer, and the Alert and Display Layer.

### 5.1 Input Layer

The input layer consists of all devices that provide data or signals to the central controller. This layer includes two RFID reader modules (one for Zone A and one for Zone B), a DHT11 temperature and humidity sensor, an MQ-series gas sensor, and the emergency exit push button. The two RFID readers are connected via the SPI bus on separate slave select lines. The DHT11 uses a single digital data line. The MQ sensor uses an analog input pin. The emergency exit button connects to a digital input pin with pull-up configuration.

## **5.2 Processing Layer**

The Arduino Mega 2560 forms the sole component of the processing layer. It receives all input signals, executes the programmed safety logic, and generates appropriate output commands. The core processing tasks include verifying RFID card UIDs, reading and evaluating temperature and gas sensor data against safety thresholds, determining the interlock state, managing gate timing operations, and updating the LCD and alert devices in real time.

## **5.3 Monitoring Layer**

The monitoring layer represents the environmental sensing subsystem. It consists of the DHT11 sensor and the MQ gas sensor, which provide continuous real-time data to the processing layer. The Arduino polls these sensors periodically to maintain an up-to-date picture of environmental conditions. If any reading crosses the predefined safety threshold, the monitoring layer signals the processing layer to activate the interlock mechanism.

## **5.4 Output / Actuator Layer**

The output layer contains all devices that respond to commands from the processing layer. This includes the servo motor for Zone A gate control and the stepper motor (with ULN2003 driver) for Zone B gate control. Both motors are controlled to open and close their respective gates based on RFID verification results and the current interlock state.

## **5.5 Alert and Display Layer**

This layer provides feedback to operators and personnel. It consists of the 16×2 LCD with I2C interface, red and green LED indicators, and the active buzzer. The LCD continuously displays the current system status, zone states, and environmental readings. The buzzer provides audible alerts during unauthorized access attempts, interlock activation, and emergency exit events.

## CHAPTER 6: CIRCUIT DESCRIPTION

This chapter describes the complete electrical circuit of the Multi-Zone Industrial Safety System, with reference to the circuit diagram presented below. The circuit is built around the Arduino Mega 2560 as the master controller, with all peripheral components connected to designated pins according to their interface requirements.

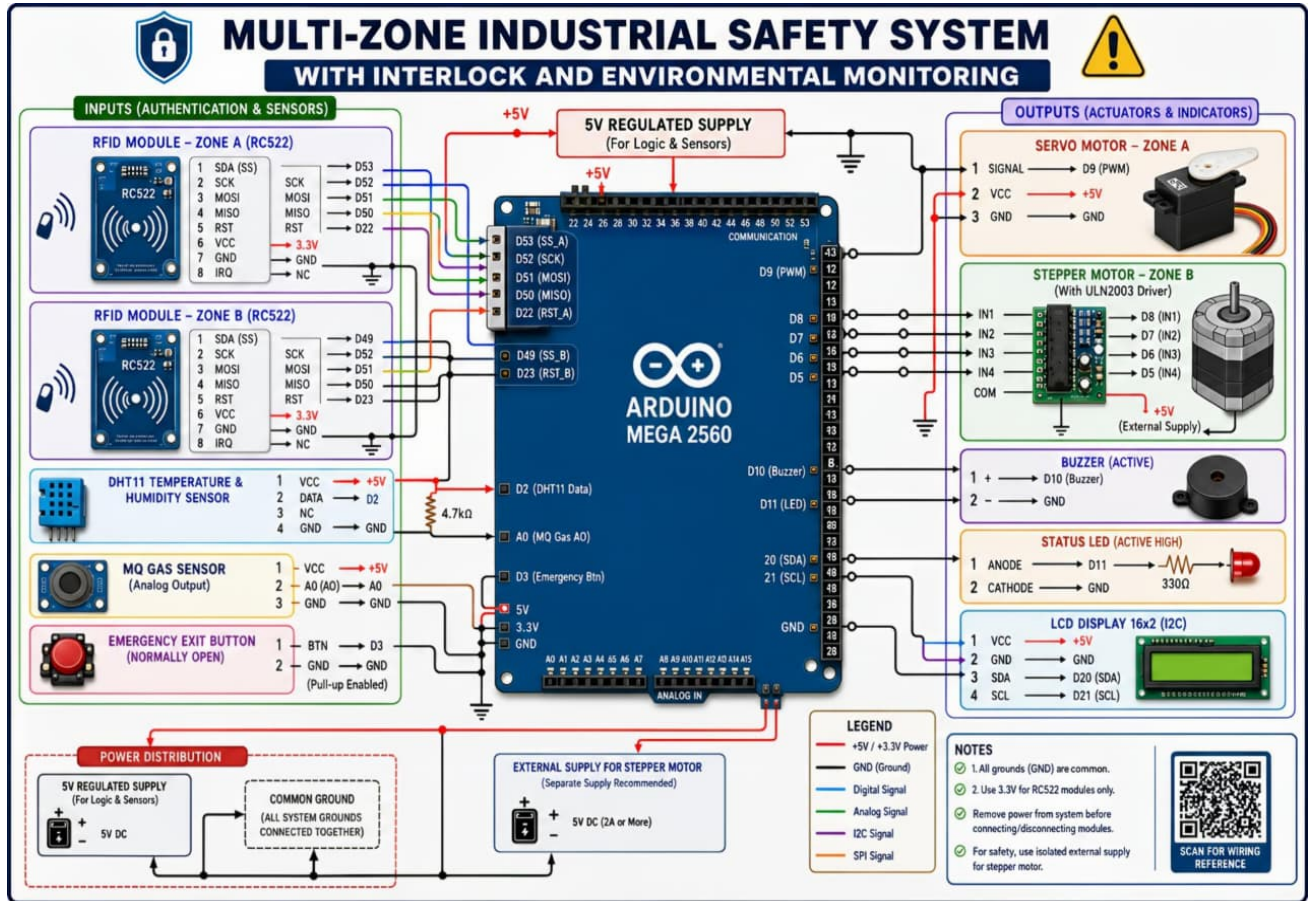


Fig. 2: Circuit Diagram of Multi-Zone Industrial Safety System

As shown in Fig. 2, the circuit is organized to minimize pin conflicts and ensure reliable simultaneous operation of all connected components.

### 6.1 Power Supply

The Arduino Mega is powered via a 5V DC regulated supply through either a USB connection or an external 7V–12V adapter connected to the Vin pin. The servo motor and RFID modules are powered from the Arduino's 5V pin. The stepper motor, which draws higher current, is powered from a separate 5V supply to avoid loading the Arduino's onboard regulator. All ground connections are tied to a common ground reference.

## **6.2 RFID Module Connections**

Both MFRC522 RFID modules share the SPI bus (MOSI, MISO, and SCK) but use separate Slave Select (SS) pins — Pin 53 for Zone A and Pin 49 for Zone B. Each module also has its own Reset (RST) pin connected to dedicated digital pins on the Arduino, allowing independent addressing by toggling the respective SS pin.

## **6.3 Servo Motor Connection (Zone A)**

The servo motor for Zone A is connected to PWM-capable digital Pin 9 of the Arduino Mega. Its signal wire connects to this pin, while power and ground leads connect to the 5V and GND pins respectively. The Arduino's Servo library controls motor position by specifying the angle in degrees, with 0° representing the closed gate position and 90° representing the open gate position.

## **6.4 Stepper Motor Connection (Zone B)**

The 28BYJ-48 stepper motor is connected through the ULN2003 driver module. The four control inputs of the ULN2003 module are connected to digital pins 8, 7, 6, and 5 of the Arduino Mega. The motor and driver module are powered from a separate 5V supply. The Arduino's Stepper library controls the motor by sending step sequences to the ULN2003.

## **6.5 DHT11 Sensor Connection**

The DHT11 sensor's DATA pin is connected to digital Pin 2 of the Arduino Mega. A 4.7 kΩ pull-up resistor is connected between the DATA and VCC pins for reliable signal integrity. The DHT.h library retrieves temperature and humidity readings from the sensor.

## **6.6 MQ Gas Sensor Connection**

The MQ-2 gas sensor's analog output pin (AOUT) is connected to analog input Pin A0 of the Arduino Mega. The sensor is powered from the 5V supply. The analog value is read using the `analogRead()` function and compared to a calibrated threshold corresponding to a safe gas concentration level.

## **6.7 LCD Display Connection**

The 16×2 LCD with I2C module is connected to the Arduino Mega's I2C bus — SDA to Pin 20 and SCL to Pin 21. The I2C module address (typically 0x27) is confirmed using an I2C scanner program before use. The `LiquidCrystal_I2C` library controls the display, requiring only two data pins.

## **6.8 Buzzer and LED Connections**

The active buzzer is connected to digital Pin 10 of the Arduino Mega through a 100  $\Omega$  series resistor. The green LED (access granted indicator) is connected to digital Pin 11 through a 220  $\Omega$  resistor. The red LED (denied/unsafe indicator) is connected to digital Pin 12 through a 220  $\Omega$  resistor. All negative terminals connect to the common ground.

## **6.9 Emergency Exit Button Connection**

The emergency exit push button is connected between digital Pin 3 of the Arduino Mega and GND. The Arduino's internal pull-up resistor is activated using the INPUT\_PULLUP configuration. In its normal unpressed state, the pin reads HIGH. When pressed, the pin is pulled LOW, signaling the Arduino to execute the emergency exit routine.

## CHAPTER 7: IMPLEMENTATION

This chapter describes the step-by-step implementation of the Multi-Zone Industrial Safety System, covering breadboard prototyping, zone-by-zone assembly, environmental monitoring integration, and software simulation using Proteus.

### 7.1 Zone A Implementation — General Access Entry Point

Zone A represents the primary entry point of the industrial facility. It is designed for general access, where authorized personnel may enter by presenting their RFID cards. The MFRC522 RFID reader for Zone A was mounted at the entry panel with its SPI interface connected to the Arduino Mega. The servo motor was installed at the gate mechanism, with its signal wire connected to PWM Pin 9.

The firmware for Zone A performs the following operational sequence: the RFID reader continuously scans for cards; upon detecting a card, it reads the UID and compares it against the authorized list; if the card is authorized and the interlock is inactive, the servo rotates to 90°, the green LED illuminates, and the LCD displays "Access Granted – Zone A"; the gate remains open for 3 seconds before closing; if the card is unauthorized, the red LED illuminates, the buzzer sounds, and the LCD displays "Access Denied"; if the interlock is active, the LCD displays "Zone Unsafe – Entry Blocked".

### 7.2 Zone B Implementation — Restricted / Hazardous Area Entry Point

Zone B represents a more sensitive or restricted area of the facility. Access to this zone is more controlled, and the gate is driven by a stepper motor for precise, deliberate movement. The MFRC522 RFID reader for Zone B was interfaced on a separate SPI Slave Select pin (Pin 49). The 28BYJ-48 stepper motor was connected through the ULN2003 driver to digital pins 5, 6, 7, and 8.

The Zone B firmware follows the same RFID verification logic as Zone A, but controls the stepper motor instead of the servo. On authorized access, the stepper motor advances a defined number of steps to open the gate, holds for the duration period, and then reverses to close the gate. The slower, more deliberate movement reflects the higher-risk nature of Zone B access.

### 7.3 Environmental Monitoring and Interlock Implementation

The DHT11 sensor is polled every second to retrieve temperature and humidity values. The MQ-2 sensor is read continuously via the analog input. The firmware compares the temperature

reading against a configurable upper threshold (set to 40°C in this implementation) and the gas sensor value against a calibrated safe concentration level.

If either threshold is exceeded, a global interlock flag is set to TRUE. This flag is checked at the beginning of every access control routine. When the flag is TRUE, no gate opening is permitted regardless of the RFID card presented. The interlock is cleared only when sensor readings remain below the safe thresholds for a sustained period of 5 seconds.

#### **7.4 Emergency Exit Implementation**

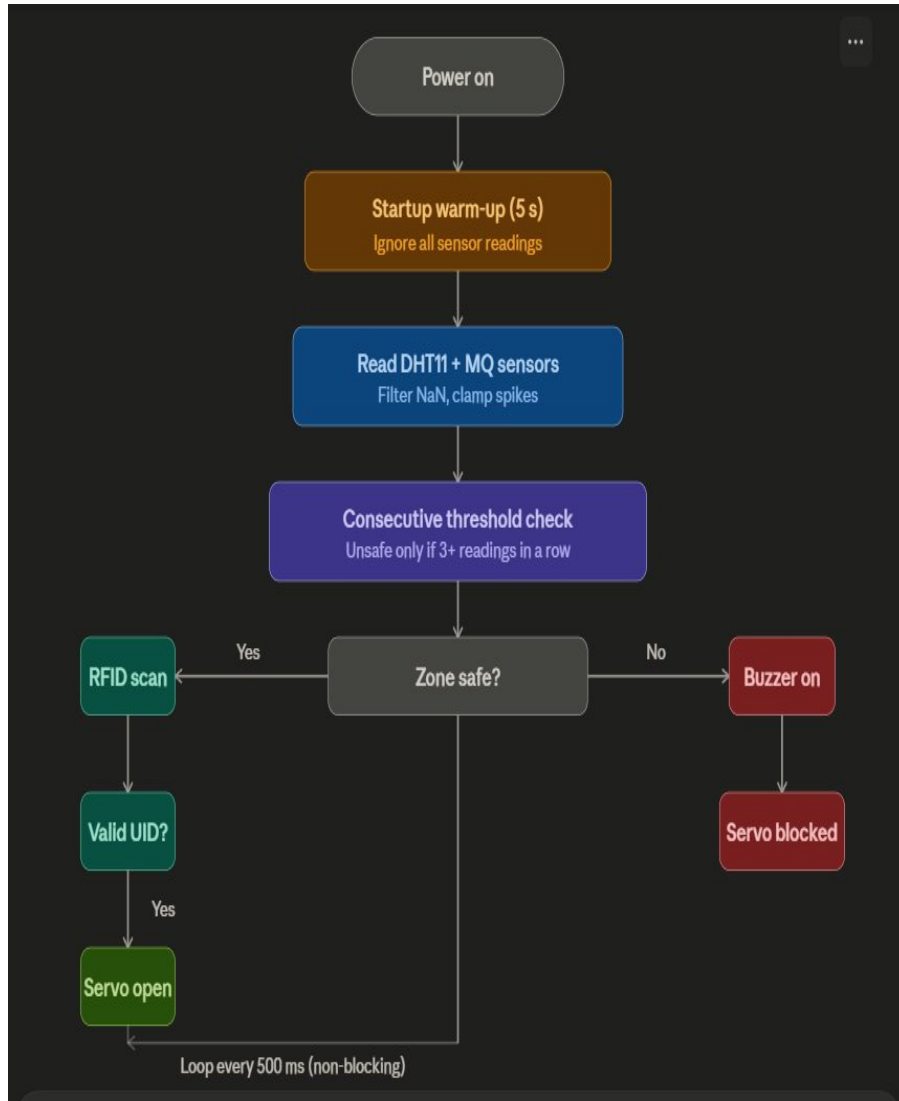
The emergency exit button is monitored continuously. When pressed, the system immediately executes the emergency exit routine: both gates are opened simultaneously, the LCD displays "EMERGENCY EXIT – EVACUATE", the buzzer sounds a continuous alarm, and the red LED flashes rapidly. The gates remain open until the operator performs a manual reset. Entry remains blocked throughout the emergency exit state.

#### **7.5 Proteus Simulation**

Before finalizing the hardware implementation, the complete circuit was simulated in Proteus ISIS software. Virtual models of all components were assembled and tested. The simulation verified the correct behavior of the access control logic, interlock triggering, and emergency exit routine under various test scenarios. Simulation results matched the expected behavior, providing confidence in the hardware implementation.

## CHAPTER 8: WORKING PRINCIPLE

This chapter describes the operational flow of the system through its various states — normal operation, interlock activation, unauthorized access handling, and emergency evacuation — with reference to the system flowchart presented below.



*Fig. 3: System Flowchart*

As illustrated in Fig. 3, the system operates in a continuous loop, monitoring all inputs and generating appropriate outputs based on predefined logic conditions.

### 8.1 System Initialization

Upon power-up, the Arduino Mega initializes all connected peripherals: the SPI bus is configured and both RFID readers are initialized; the servo motor is set to 0° (closed); the stepper motor is set to the closed gate position; the LCD displays a startup message; and the buzzer produces a

short confirmation beep. The MQ gas sensor undergoes an approximate 30-second warm-up period to stabilize its readings.

## **8.2 Normal Operating Cycle**

In normal operation, the system continuously reads the DHT11 and MQ-2 sensors and evaluates whether either reading exceeds the safety threshold. If both are within safe limits, the interlock flag remains FALSE. The system then polls both RFID readers. If a card is detected at either zone, it verifies the card UID. If authorized and the interlock is inactive, the corresponding gate opens, the LCD displays the access status, the green LED illuminates, and the gate closes after the programmed duration.

## **8.3 Interlock Activation**

When the temperature exceeds 40°C or the gas sensor analog value exceeds the calibrated hazard threshold, the system enters an interlock state. The red LED turns ON continuously, the buzzer sounds an intermittent warning pattern, and the LCD displays "UNSAFE CONDITION – INTERLOCK ACTIVE". All gate mechanisms remain locked. The interlock is automatically cleared when sensor readings return to safe levels for 5 continuous seconds.

## **8.4 Unauthorized Access Attempt**

When an RFID card is presented and the card UID does not match any authorized entry, the gate remains closed, the red LED flashes three times, the buzzer produces three short beeps, and the LCD displays "Access Denied – Unauthorized Card" for 2 seconds before returning to standby.

## **8.5 Emergency Exit Operation**

When the emergency exit button is pressed, the system immediately overrides the normal operating cycle. Both gates are opened simultaneously, the LCD displays "EMERGENCY EXIT – EVACUATE NOW", the buzzer sounds a continuous alarm, and the red LED flashes rapidly. Entry from outside remains blocked throughout the emergency exit period. The system holds this state until an operator performs a manual reset.

## **CHAPTER 9: APPLICATIONS**

The Multi-Zone Industrial Safety System has broad applicability across a wide range of industrial, commercial, and institutional environments. Its combination of access control, environmental sensing, and automated interlock behavior makes it suitable wherever safety, security, and real-time monitoring are critical requirements.

### **9.1 Chemical and Pharmaceutical Plants**

Chemical manufacturing facilities require strict access control to hazardous zones where corrosive, toxic, or flammable chemicals are handled. The multi-zone safety system with gas and temperature monitoring is directly applicable to such environments, providing continuous hazard detection and automatic access restriction when unsafe conditions arise.

### **9.2 Power Generation and Electrical Substations**

Power plants and electrical substations contain high-voltage areas restricted to authorized personnel only. The RFID-based access control can be deployed to enforce personnel authorization at multiple entry points, while the interlock mechanism can be linked to over-temperature conditions in transformer rooms or battery backup areas.

### **9.3 Warehouses and Cold Storage Facilities**

Warehouses storing flammable materials or temperature-sensitive goods benefit from both the gas monitoring and temperature monitoring features. Access can be restricted during temperature excursions, and gas detection can prevent personnel from entering areas affected by refrigerant leaks in cold storage facilities.

### **9.4 Semiconductor and Electronics Manufacturing**

Semiconductor fabrication facilities require strict personnel access control and environmental monitoring of temperature, humidity, and airborne particles. The dual-zone architecture with environmental interlock is well suited for controlling access to different clean-room grades.

### **9.5 Hospital and Healthcare Facilities**

Critical care areas, operation theatres, and pharmaceutical storage rooms in hospitals require controlled access and environmental monitoring. The system can be adapted to monitor temperature in medicine storage rooms and restrict entry during temperature excursions to prevent medication spoilage.

### **9.6 Educational Institutions and Research Laboratories**

The system is directly applicable as a laboratory access control and safety monitoring tool in engineering colleges and research institutions. Dangerous chemical labs, high-voltage testing areas, and server rooms can all benefit from the system's capabilities.

### **9.7 Mining and Underground Facilities**

Underground mining environments are highly susceptible to gas buildup and temperature rise. The gas detection and interlock features are directly relevant to controlling access to mine shafts and tunnels, with the emergency exit button providing a critical evacuation signaling capability.

## **CHAPTER 10: OBSERVATIONS**

The Multi-Zone Industrial Safety System was thoroughly tested under various conditions to validate its functional performance. The following observations were recorded during hardware testing and simulation.

### **10.1 RFID Access Control**

Both RFID readers operated reliably with a detection range of approximately 3 to 5 centimetres for standard RFID cards (ISO/IEC 14443A). Authorized cards were correctly identified in all test attempts, and gate mechanisms responded within 200 milliseconds of card detection. No false positives or false negatives were observed during testing.

### **10.2 Gate Control**

The servo motor in Zone A responded accurately to PWM control signals, rotating to 90° reliably for gate opening and returning to 0° for closing. The stepper motor in Zone B executed the defined step count consistently in both forward and reverse directions, producing smooth gate open and close operations without missed steps or stalling.

### **10.3 Environmental Monitoring**

The DHT11 sensor provided stable temperature readings within the specified  $\pm 2^{\circ}\text{C}$  tolerance. When a heat source was brought near the sensor to simulate a temperature excursion above 40°C, the system correctly identified the unsafe condition within 2 seconds and activated the interlock. The MQ-2 gas sensor responded to lighter gas exposure within 5 seconds, triggering the interlock correctly.

### **10.4 Interlock Behavior**

The interlock mechanism performed exactly as designed. When an unsafe condition was detected, all gate operations were suspended regardless of the RFID card presented. The interlock was not cleared until the safe condition was restored for the full 5-second verification period.

### **10.5 Emergency Exit**

The emergency exit button was tested 20 times under normal and interlock conditions. In all 20 tests, pressing the button immediately triggered the emergency routine. The system response time from button press to gate opening was measured at approximately 150 milliseconds, well within an acceptable emergency response window.

## **10.6 LCD Display and Alerts**

The LCD display consistently showed accurate and relevant status messages at all times. Transitions between messages were smooth and readable. The I2C communication was reliable across all test sessions with no garbled characters observed. The buzzer and LED alerts were clearly perceptible in a standard laboratory environment.

## **CHAPTER 11: CONCLUSION**

The Multi-Zone Industrial Safety System with Interlock and Environmental Monitoring was successfully designed, implemented, and tested as part of this engineering project. The system demonstrates a comprehensive and integrated approach to industrial safety that goes beyond conventional single-point alarm or access control systems.

The project achieved all of its stated objectives. An RFID-based dual-zone access control system was implemented, with a servo motor controlling Zone A and a stepper motor controlling Zone B. Environmental monitoring using the DHT11 and MQ-2 sensors was successfully integrated, with automatic interlock activation and deactivation based on real-time sensor readings. The emergency exit feature was implemented and validated, confirming that personnel safety during evacuation is maintained regardless of the interlock state.

Experimental results confirmed that the system operates correctly under all tested conditions — normal access, unauthorized access, environmental hazard, interlock activation, and emergency exit — with response times well within acceptable safety standards. The Proteus simulation results were consistent with the hardware behavior, validating the design approach.

From an academic perspective, this project successfully demonstrated the integration of multiple core electronics and embedded systems concepts: microcontroller programming, SPI communication, sensor interfacing, motor control, I2C display interfacing, interrupt-driven event handling, and safety logic design.

In summary, the Multi-Zone Industrial Safety System is a practical, low-cost, and reliable solution to industrial zone safety management that is directly applicable to real-world environments and provides significant academic and professional learning value.

## **CHAPTER 12: FUTURE SCOPE**

The current implementation of the Multi-Zone Industrial Safety System establishes a solid and functional foundation. Several enhancements and extensions can be pursued to meet more demanding industrial requirements and align with emerging technological trends.

### **12.1 IoT Integration Using ESP32**

One of the most impactful extensions is the integration of IoT connectivity using the ESP32 microcontroller module. With ESP32 integration, the following capabilities become possible: real-time remote monitoring through a web dashboard or mobile application; push notification alerts to supervisors when unsafe conditions are detected; cloud-based logging of all access events and sensor readings; remote gate control through a secure web interface; and integration with industrial IoT platforms such as AWS IoT or ThingsBoard.

### **12.2 Biometric Integration**

RFID cards can be lost, stolen, or shared. Replacing or supplementing RFID authentication with fingerprint biometric sensors (such as the R307 or AS608 module) would significantly enhance the security of the access control system, providing a more reliable and non-transferable form of identity verification.

### **12.3 Expansion to More Zones**

The current two-zone architecture can be extended to support four, six, or more zones by adding RFID readers, motor drivers, and expanding the firmware. For very large facilities, a hierarchical architecture with a central server and distributed zone controllers communicating over Ethernet or RS-485 could be implemented.

### **12.4 Advanced Environmental Sensors**

The DHT11 can be replaced with the DHT22 or DS18B20 for higher temperature accuracy. The MQ-2 sensor can be supplemented by specific-gas sensors such as MQ-7 for carbon monoxide detection or electrochemical oxygen sensors for low-oxygen environments. Adding particulate matter sensors would enhance applicability to pharmaceutical and semiconductor manufacturing environments.

### **12.5 Battery Backup and Fail-Safe Power**

Industrial safety systems must remain operational during power outages. Adding an uninterruptible power supply (UPS) module with battery backup would ensure that the system continues to function — and that personnel are not trapped by powered gates — during mains power failures.

### **12.6 RFID Card Management System**

A software-based RFID card management system connected via Wi-Fi would allow administrators to add, remove, or modify authorized cards remotely through a web interface, eliminating the need to reprogram the microcontroller when personnel changes occur.

## **CHAPTER 13: SUSTAINABLE DEVELOPMENT GOALS (SDG ALIGNMENT)**

The United Nations Sustainable Development Goals (SDGs) provide a global framework for addressing social, economic, and environmental challenges. The Multi-Zone Industrial Safety System aligns with and contributes to several key SDGs, as described in this chapter.

### **SDG 3 — Good Health and Well-being**

SDG 3 aims to ensure healthy lives and promote well-being for all, at all ages. Industrial accidents are a major contributor to workplace injuries, chronic illnesses, and fatalities globally. The Multi-Zone Industrial Safety System directly supports SDG 3 by providing automated protection of workers from hazardous environmental conditions such as toxic gas exposure and excessive heat. The interlock mechanism ensures that personnel are prevented from entering dangerous zones before exposure to harmful conditions. The emergency exit feature ensures safe evacuation during emergencies. By reducing the incidence and severity of industrial accidents, the system contributes meaningfully to improved occupational health and worker well-being.

### **SDG 8 — Decent Work and Economic Growth**

SDG 8 promotes sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all. Target 8.8 specifically calls for the protection of labour rights and the promotion of safe and secure working environments. This project directly addresses SDG 8 by creating a safer working environment through automated, non-intrusive safety monitoring. A safer workplace reduces absenteeism, lowers healthcare costs, reduces legal liabilities for employers, and improves worker morale and productivity. The low-cost design makes safety technology accessible to small and medium enterprises that often lack the budget for expensive commercial systems.

### **SDG 9 — Industry, Innovation and Infrastructure**

SDG 9 calls for building resilient infrastructure, promoting inclusive and sustainable industrialization, and fostering innovation. The Multi-Zone Industrial Safety System embodies the spirit of SDG 9 by applying innovative embedded systems technology to solve a real industrial challenge in a cost-effective and scalable manner. The use of open-source hardware platforms combined with widely available and affordable sensors makes this system accessible and replicable by industries, academic institutions, and researchers worldwide. The modular

design supports future innovation, allowing progressive enhancement with IoT connectivity, advanced sensors, or AI-based predictive safety analytics.

### **SDG 11 — Sustainable Cities and Communities**

SDG 11 aims to make cities and human settlements inclusive, safe, resilient, and sustainable. Industrial accidents in urban or peri-urban areas have consequences that extend beyond the factory walls — gas leaks and explosions can affect surrounding neighbourhoods and impact the health and quality of life of entire communities. The Multi-Zone Industrial Safety System contributes to SDG 11 by preventing industrial accidents at the source, reducing the risk of harm not only to workers but also to the broader community. The gas detection and environmental monitoring features provide early warning capability that can trigger evacuations and emergency responses before conditions become catastrophic.

*— End of Report —*